

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I A Sai Swetha(192319072) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

1(a) Full Citation and Application Area

Full Citation

Ismail, L., Materwala, H., Tayefi, M., Ngo, P., & Karduck, A. P. **“Type 2 Diabetes with Artificial Intelligence Machine Learning: Methods and Evaluation.”** *Archives of Computational Methods in Engineering*, Volume 29, pages 313–333, 2022. DOI: **10.1007/s11831-021-09582-x**.

Application Area

The article belongs to:

- Artificial Intelligence
- Machine Learning
- Healthcare Informatics
- Medical Diagnosis
- Diabetes Risk Prediction
- Clinical Decision Support

The authors explain that Type 2 diabetes is associated with several interacting risk factors, including lifestyle, medical, demographic, psychosocial and hereditary factors. Early prediction can support diagnosis and prevention planning.

Specific Problem Addressed

The main problem is:

How can different machine-learning classification algorithms be objectively compared for predicting Type 2 diabetes when previous studies use different datasets and evaluation methods?

The authors observed that previous research used different datasets and different performance measures. As a result, directly comparing the reported results was difficult.

The article attempts to solve this problem by creating a **unified experimental framework**.

The authors evaluated:

- 35 machine-learning algorithms
- 3 real-world diabetes datasets
- 9 feature-selection algorithms
- Models with and without feature selection
- Accuracy
- F-measure
- Execution time

This makes the article especially relevant to the **CO4 topic of classification and decision-making**.

1(b) Key Objective of the Article

150–200 Word Summary

The main objective of the article is to provide a systematic and unified evaluation of machine-learning algorithms for Type 2 diabetes prediction. The authors identified a major research gap in earlier studies: different researchers used different datasets, algorithms and evaluation metrics, making it difficult to determine which algorithms were actually more effective. To address this problem, the authors evaluated 35 machine-learning algorithms using three real-life diabetes datasets: the PIMA Indian dataset, the UCI dataset and the MIMIC III dataset. The algorithms were evaluated both with and without feature selection. Nine feature-selection methods were used to identify important diabetes risk factors. The models were compared mainly using accuracy, F-measure and execution time. The study found that model performance was strongly dependent on the characteristics of the dataset. Bagging with Logistic Regression performed best for the balanced dataset, while Random Forest performed particularly well on imbalanced datasets. The authors also identified important diabetes predictors including age, ethnicity, family history, hypertension, obesity, cardiovascular disease and cholesterol.

Research Gap

The research gap is one of the most important aspects of this article.

Previous studies had already applied many machine-learning algorithms to diabetes prediction.

However, the authors noticed that:

Different researchers
↓
Different datasets
↓
Different preprocessing
↓
Different evaluation metrics
↓
Difficult to compare results

For example, one study might report accuracy using one dataset, while another might report F-measure using a different dataset.

Therefore, saying that one algorithm is better than another becomes difficult.

How the authors addressed the gap

The authors used:

```
3 real-world datasets
      +
35 ML algorithms
      +
9 feature-selection algorithms
      +
Common evaluation procedure
      ↓
Unified comparison
```

The authors explicitly state that their objective was to address the lack of objective comparison using unified datasets and evaluation measures.

TASK 2: METHODOLOGY ANALYSIS

2(a) ML Techniques Used

The article evaluates a very large number of machine-learning techniques rather than relying on a single model.

These include:

- Decision Tree
- Bayesian Network
- Naïve Bayes
- K-Nearest Neighbour
- K-Star
- Logistic Regression
- Support Vector Machine
- Artificial Neural Network
- ZeroR
- OneR
- JRip
- Decision Table
- Random Tree
- Random Forest
- REPTree
- K-Means combined with classification
- Bagging

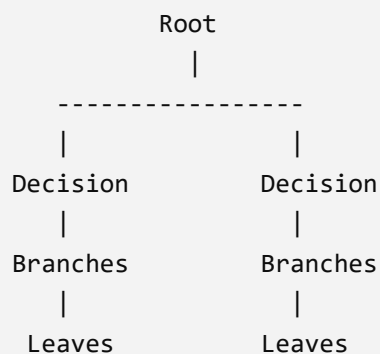
- Boosting
- Stacking

The article describes Decision Trees as models that construct sequences of decisions and outcomes, selecting branches according to information gain.

Decision Tree

A Decision Tree is a supervised learning algorithm.

Its structure contains:



The article uses **Information Gain** for selecting decision-tree splits.

The basic equation is:

where:

- = risk factor
- = entropy before splitting
- = conditional entropy after splitting

The risk factor producing the largest information gain is preferred for the split.

Naïve Bayes

Naïve Bayes is based on Bayes' theorem.

represents the probability of a diabetes class given the observed risk factors.

The model assumes that the risk factors are conditionally independent given the class.

The article uses Naïve Bayes as one of the statistical/probabilistic classification techniques.

Logistic Regression

Logistic Regression predicts the probability that a patient belongs to the diabetes class.

The article represents the probability using a sigmoid/logistic function:

where:

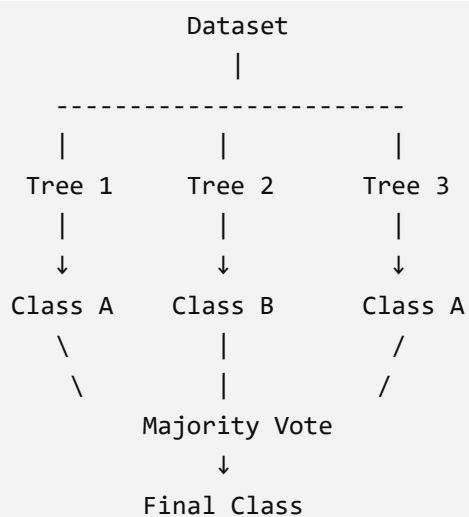
- = intercept
- = regression coefficient
- = risk factor

The coefficients are estimated using maximum likelihood.

Random Forest

Random Forest is an ensemble method consisting of multiple decision trees.

Conceptually:



The article found Random Forest particularly effective for imbalanced datasets because its bootstrap samples can provide balanced subsets for individual trees.

Datasets Used

The study used three real-world datasets:

1. PIMA Indian Dataset

The PIMA Indian dataset originates from the **National Institute of Diabetes and Digestive and Kidney Diseases**.

It is used to predict whether a patient has diabetes based on diagnostic measurements.

2. UCI Dataset

The UCI dataset contains patient and hospital outcome information from **130 US hospitals between 1999 and 2008**.

The dataset includes variables such as:

- Race
- Gender
- Age
- Diagnosis 1
- Diagnosis 2
- Diagnosis 3
- Diabetes medication

The researchers derived a diabetes class label from the diabetes-medication information and converted certain diagnosis information into binary risk-factor features.

3. MIMIC III

MIMIC III is a large critical-care database containing information from **more than 40,000 patients** admitted to critical-care units at Beth Israel Deaconess Medical Center between 2001 and 2012.

The study used available diabetes-risk features including:

- Ethnicity
- Gender
- Age
- Diabetes status
-

Dataset Preparation

The authors performed:

1. Feature extraction

- 2. Feature selection
- 3. Classification
- 4. Ten-fold cross-validation
- 5. Repeated runs
- 6. Performance averaging

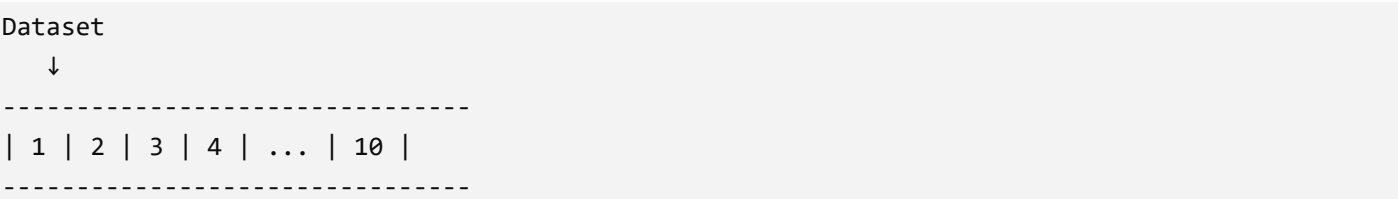
For feature selection, the researchers used nine feature-selection algorithms.

Features appearing in more than 50% of the selection algorithms were selected as important features.

Ten-Fold Cross-Validation

The authors used **10-fold cross-validation**.

The dataset is divided into ten parts.



One part is used for testing while the remaining nine parts are used for training.

This process is repeated until each partition has served as the test set.

Finally, the results are averaged.

This provides a more robust estimate than relying on a single train-test split.

2(b) Mapping to CO4

The article strongly relates to the CO4 syllabus.

CO4 Topic	Connection to Article
Inductive Learning	Models learn diabetes patterns from labelled patient data
Decision Trees	Decision Tree and REPTree are evaluated

CO4 Topic	Connection to Article
Statistical Learning	Naïve Bayes and Logistic Regression are evaluated
Classification	Diabetes vs non-diabetes classification
Decision-making	Models support diabetes risk prediction
Feature Selection	Nine feature-selection methods are evaluated

The assessment requires an article related to Inductive Learning, Decision Trees, Statistical Learning or Reinforcement Learning, and this article directly covers several of these categories.

CO4-AT3

Methodological Strength

Strength: Unified Comparison

The strongest aspect of the methodology is its unified experimental setup.

Instead of comparing unrelated results from different studies, the researchers evaluated many algorithms using common datasets and evaluation procedures.

This makes the comparison more meaningful.

Another strength is the use of **three different datasets**, including a large critical-care dataset.

The use of ten-fold cross-validation and repeated runs also improves the reliability of the reported comparisons.

Methodological Limitation

Limitation: Dataset Imbalance

A major limitation is class imbalance.

The UCI and MIMIC III datasets contain strongly imbalanced diabetes classes. The authors found that many algorithms predicted the majority class and failed to detect the minority class.

This is particularly important in healthcare.

Suppose:

950 non-diabetic
50 diabetic

A model that predicts everyone as non-diabetic obtains:

But it identifies:

of diabetic patients.

Therefore, accuracy alone can be misleading.

The authors themselves emphasise the importance of using F-measure along with accuracy for imbalanced healthcare data.

TASK 3: RESULTS & CRITICAL EVALUATION

3(a) Key Results

The paper compared algorithms using:

- Accuracy
- F-measure
- Execution time

The authors did not rely only on accuracy because class imbalance can make accuracy misleading.

Main Findings

Dataset / Condition	Best-performing approach reported
PIMA Indian	Bagging + Logistic Regression with feature selection
UCI	Random Forest
MIMIC III	Random Forest

Dataset / Condition	Best-performing approach reported
Balanced data	Bagging-LR performs best
Imbalanced data	Random Forest performs best

The authors conclude that **Bagging-LR was most accurate for balanced data**, whereas **Random Forest was most accurate for imbalanced data**.

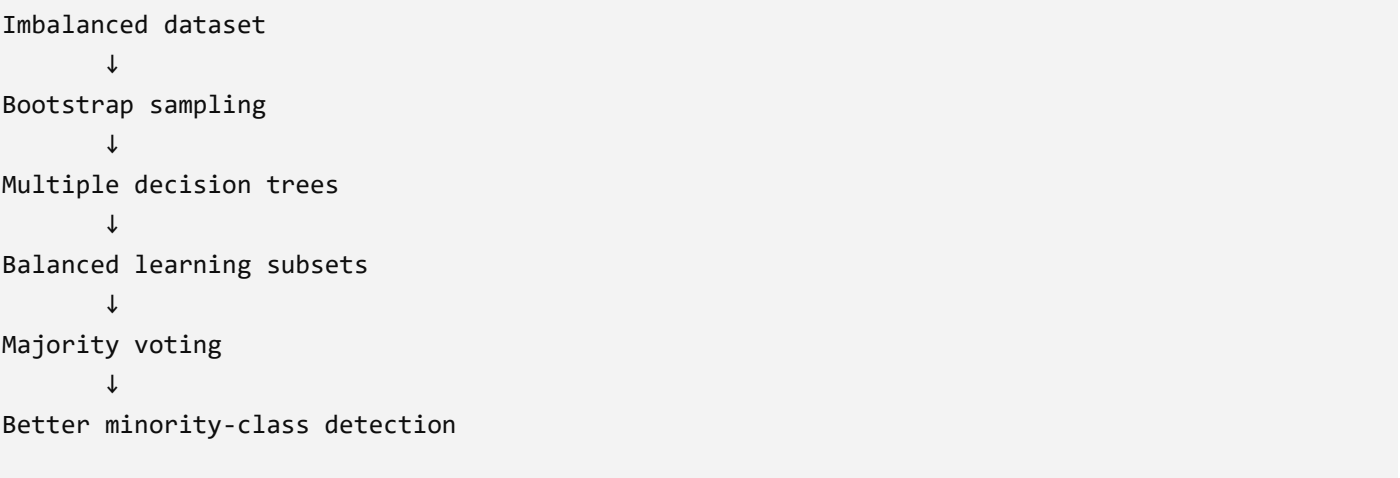
Why Random Forest Performed Well

Random Forest creates multiple decision trees using bootstrap samples.

For an imbalanced dataset, its sampling mechanism can provide each tree with a more balanced representation of classes.

The paper explains that Random Forest performs better because each tree uses a bootstrap sample containing minority-class observations and a similarly sized sample from the majority class.

Therefore:



Feature Selection Results

The study identified several significant risk factors.

The important factors included:

1. **Age**
2. **Ethnicity**
3. **Family history of diabetes**
4. **Hypertension**
5. **Obesity**
6. **Cardiovascular disease**
7. **Cholesterol**

These factors belong to different categories including demographic, hereditary, medical and lifestyle-related factors.

Effect of Feature Selection

Feature selection had several benefits.

The authors found that:

- Important predictors could be identified.
- Execution time decreased.
- The number of features was reduced.
- Performance was generally maintained.

For example, the execution time for PIMA Indian was reduced by up to **2.5 times**, while UCI execution time was reduced by approximately **1.23 times** after feature selection.

This demonstrates that feature selection is useful not only for prediction but also for computational efficiency.

3(b) Critical Evaluation

Are the Metrics Realistic?

The reported results should be interpreted carefully.

The paper demonstrates that some algorithms can achieve strong predictive performance, but high accuracy does not automatically mean that a model is clinically reliable.

This is especially important because the study identifies severe class imbalance in some datasets.

For example, several algorithms predicted only the majority non-diabetic class and therefore had no meaningful F-measure for the minority class.

Therefore:

High accuracy $\neq$ good clinical model

A clinically useful model should demonstrate:

- Good sensitivity/recall
- Good specificity
- Good F1-score
- Good calibration
- Robustness across populations
- External validation

Potential Dataset Bias

The datasets originate from different healthcare contexts.

PIMA Indian

Represents a particular population and may not generalise to every demographic group.

UCI

Contains information from US hospitals.

MIMIC III

Contains critical-care patients and therefore may not represent the general population.

Consequently, a model trained on MIMIC III may not perform similarly in:

- Primary-care settings
- Community health centres
- Young populations
- Different countries
- Different ethnic populations

Therefore, external validation is necessary.

Potential Evaluation Bias

The study uses ten-fold cross-validation, which is a strength.

However, further evaluation would be useful.

For example:

1. External validation

Test the final model on a completely independent hospital dataset.

2. Prospective validation

Evaluate the model on future patient records.

3. Subgroup analysis

Measure performance separately by:

- Age
- Sex
- Ethnicity
- Hospital
- Geographic region

4. Calibration

A model should not only classify correctly but also produce reliable probabilities.

For example:

Predicted risk = 80%

should correspond approximately to an 80% observed risk in an appropriate validation population.

5. Confidence intervals

Report uncertainty around performance metrics.

Additional Experiments That Would Strengthen the Study

I would recommend:

Experiment 1: External validation on a completely independent diabetes dataset.

Experiment 2: Compare sensitivity and specificity in addition to accuracy and F-measure.

Experiment 3: Perform subgroup fairness analysis.

Experiment 4: Evaluate calibration curves.

Experiment 5: Compare traditional models with modern explainable boosting/ensemble methods.

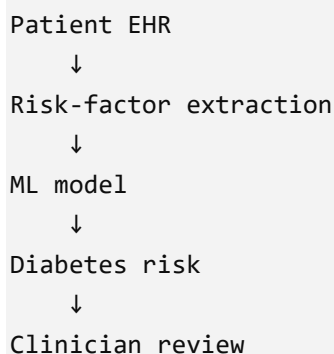
Experiment 6: Perform prospective validation using newly collected patient records.

3(c) Real-World Applicability

The techniques discussed in the article could potentially be used in:

1. Hospital Decision-Support Systems

A model could estimate diabetes risk from electronic health records.



2. Primary Healthcare

The system could help identify patients who require additional diabetes screening.

3. Population Health

Healthcare organizations could identify high-risk groups.

4. Preventive Healthcare

High-risk individuals could receive:

- Lifestyle counselling
- Follow-up testing
- Monitoring
- Preventive interventions

The article itself positions machine-learning prediction as a potential support mechanism for diagnosis, prognosis and prevention planning.

Challenges in Industry Deployment

1. Data Availability

Healthcare data may be:

- Incomplete
- Noisy
- Distributed across hospitals
- Stored in different formats

2. Data Privacy

Patient health information is highly sensitive.

Systems must comply with applicable privacy and security requirements.

3. Model Generalisation

A model trained on one hospital's data may not work equally well at another hospital.

4. Class Imbalance

Diabetes datasets may contain substantially more non-diabetic than diabetic cases.

5. Explainability

Clinicians need to understand why a model generated a particular prediction.

6. Bias

A model may perform differently for different demographic groups.

7. Computational Requirements

Large datasets such as MIMIC III require more computational resources than small datasets.

8. Clinical Validation

An ML model should undergo appropriate clinical evaluation before being used in patient care.

TASK 4: REFLECTION & LEARNING OUTCOME

4(a) Three Key Insights

Insight 1: Accuracy Alone Is Not Sufficient

The first major insight is that **accuracy can be misleading when the dataset is imbalanced**.

The article demonstrates that some models achieved apparently good accuracy while failing to identify the minority diabetes class.

Connection to CO4

This connects directly to:

Classification evaluation

and the concepts of:

- Accuracy
- Precision
- Recall
- F-measure

The article specifically recommends including F-measure when evaluating imbalanced healthcare data.

Insight 2: Dataset Characteristics Affect Model Performance

The second important insight is:

There is no universally best machine-learning algorithm.

The best algorithm depends on the dataset.

The article reports that:

- Bagging-LR performed best for balanced data.
- Random Forest performed best for imbalanced data.

Inductive learning and model selection.

The model learns patterns from a particular dataset, so changes in data distribution can affect its performance.

Insight 3: Feature Selection Improves Efficiency

The third insight is the importance of selecting meaningful features.

The article identified several important diabetes predictors, including:

- Age
- Family history
- Hypertension

- Obesity
- Cardiovascular disease
- Cholesterol
- Ethnicity

Feature selection reduced execution time while generally maintaining model performance. [pringer](#)

Connection to CO4

This connects to:

Feature selection and classification

because removing irrelevant or redundant variables can produce simpler and more efficient models.

4(b) Proposed Novel Extension

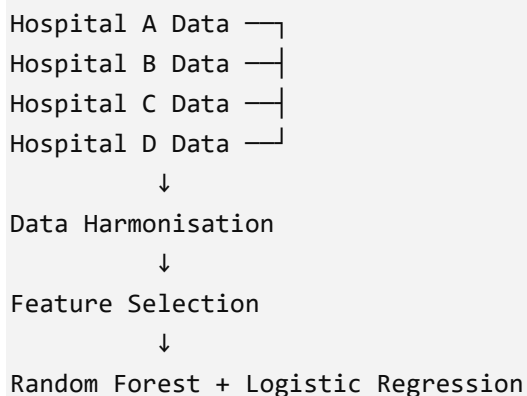
Proposed Experiment

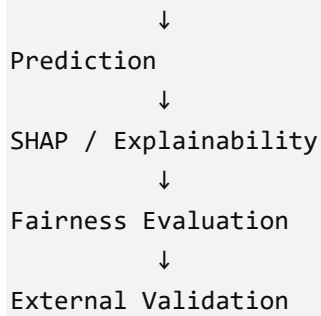
Explainable and Fair Diabetes Prediction Using Multi-Hospital Data

I would extend the research by developing a model using data from multiple hospitals and combining:

- Random Forest
- Logistic Regression
- Explainable AI
- Fairness analysis
- External validation

Proposed Architecture





Why This Extension Is Feasible

The original article already demonstrates that:

- Multiple datasets can be analysed.
- Feature selection is practical.
- Random Forest performs well on imbalanced data.
- Logistic Regression performs strongly in certain conditions.
- Weka can be used for implementation.

Therefore, extending the work toward a multi-hospital, explainable system is technically feasible.

Expected Impact

The proposed extension could provide:

Better Generalisation

Testing across hospitals would help determine whether the model works beyond a single population.

Better Fairness

Performance could be compared between demographic groups.

Better Interpretability

Explainability methods could show which factors contributed to a prediction.

Better Clinical Trust

Clinicians may be more willing to use a model if its reasoning can be examined.

Better Safety

External validation could identify situations in which the model performs poorly before deployment.

Overall Critical Review

The article is a strong contribution because it does not simply propose another diabetes prediction model. Instead, it addresses an important methodological issue: **different machine-learning studies often use different datasets and metrics, making their results difficult to compare.**

The authors address this by evaluating **35 machine-learning algorithms across three real-world datasets**, with and without feature selection.

One of the strongest findings is that **model performance depends strongly on the characteristics of the dataset**. Random Forest was particularly effective on imbalanced datasets, whereas Bagging with Logistic Regression performed best on balanced data.

Another important finding is that **accuracy should not be considered alone**. In healthcare datasets with class imbalance, a model can obtain high accuracy while failing to identify diabetic patients. The authors therefore recommend using F-measure along with accuracy.

Feature selection is another valuable contribution. The study identified important diabetes-related factors and showed that reducing the feature set could lower computational time without significantly reducing predictive performance.

However, the study could be strengthened by more extensive external validation, subgroup fairness analysis, calibration analysis, and prospective evaluation. These additions would make the findings more applicable to real clinical environments.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand